# Learning Rate Decay

Should epoch 1 and epoch 100 use the same learning rate? · Lecture 5c

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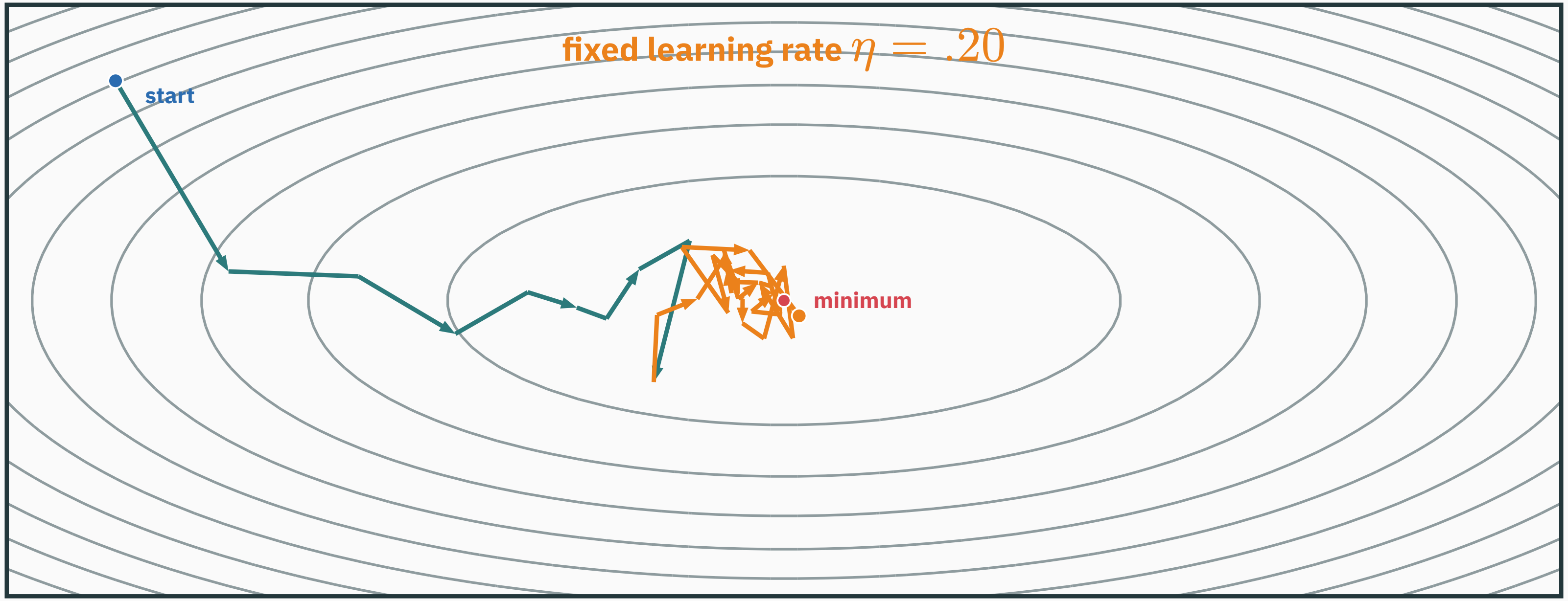
ES 667 · Deep Learning · IIT Gandhinagar

# What changes as SGD gets closer?

Q QUESTION




early updates: fast progress

later updates: motion around the minimum

# What changes as SGD gets closer?

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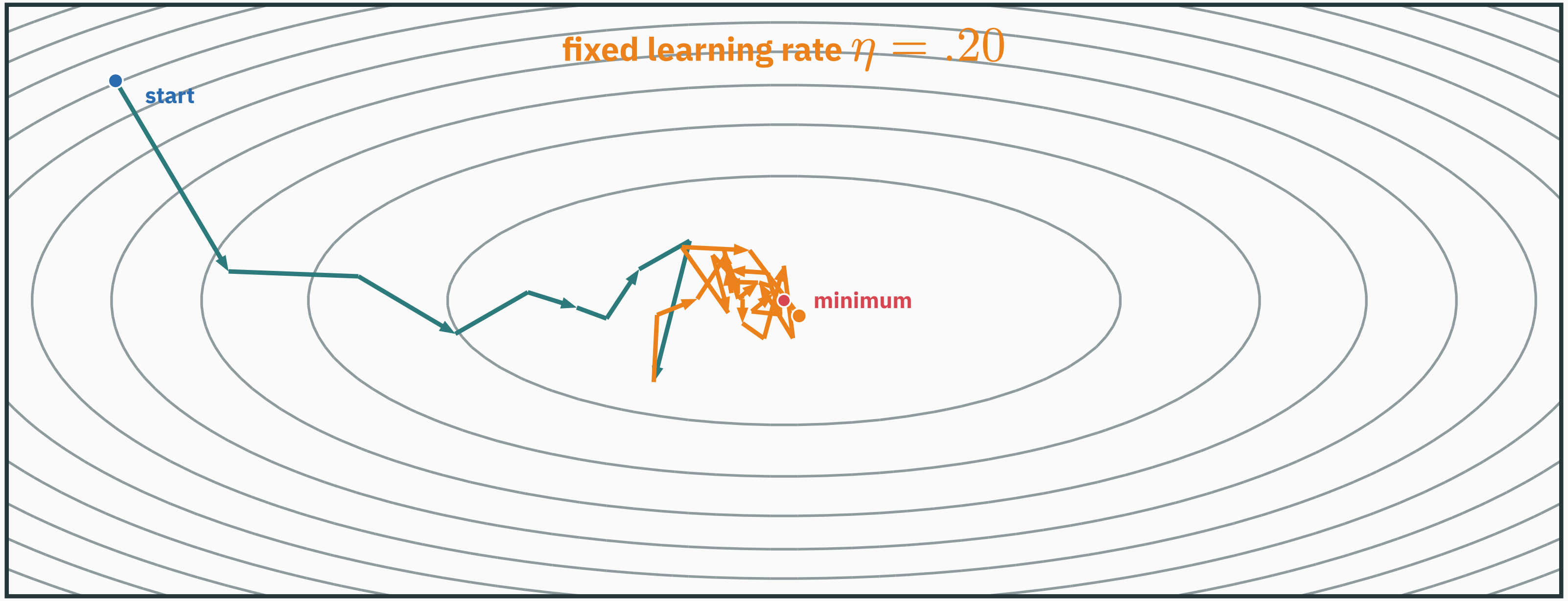


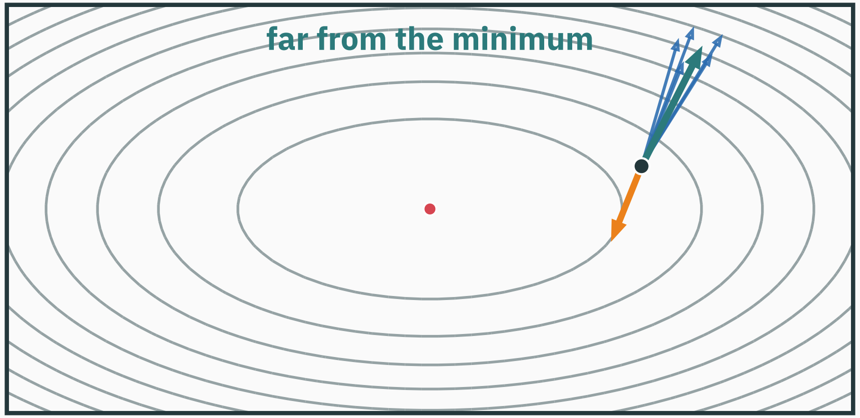

early updates: fast progress

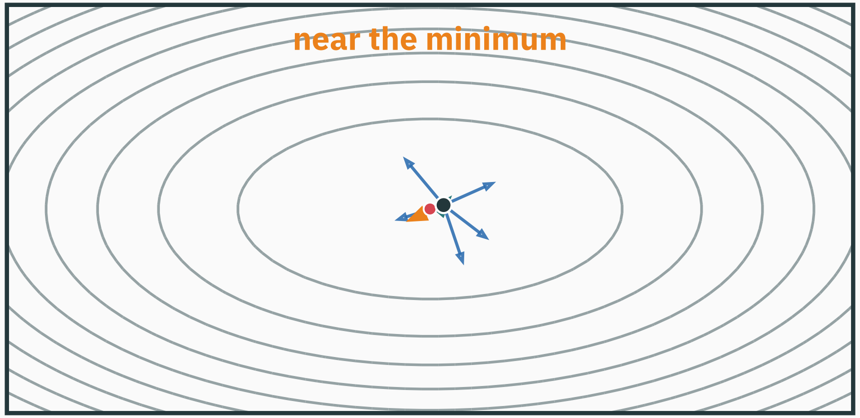
later updates: motion around the minimum

**SGD has reached the right region. The same step size now makes it bounce around.**

# A minibatch gives a noisy gradient

V VISUAL

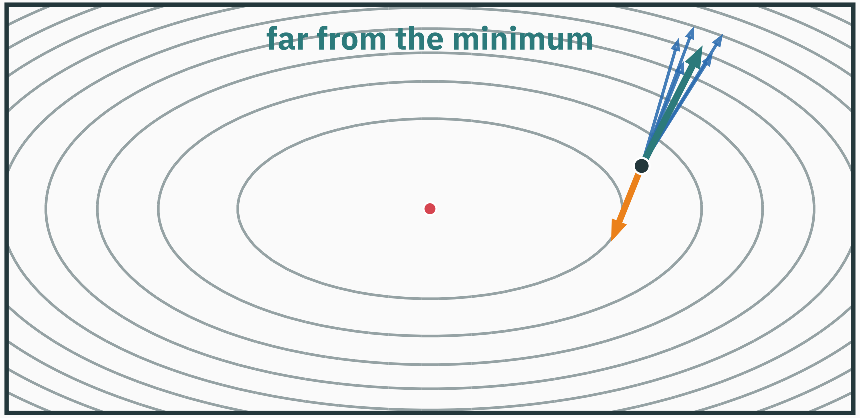


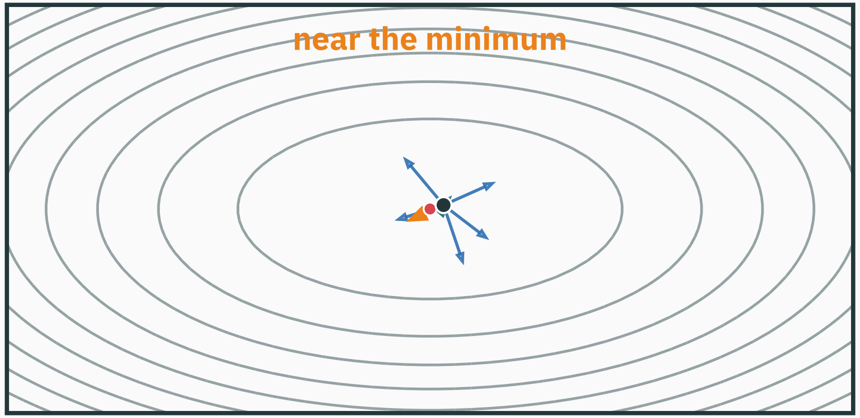




full-data gradient   minibatch gradients   chosen update

# A minibatch gives a noisy gradient

V VISUAL




full-data gradient · minibatch gradients · chosen update

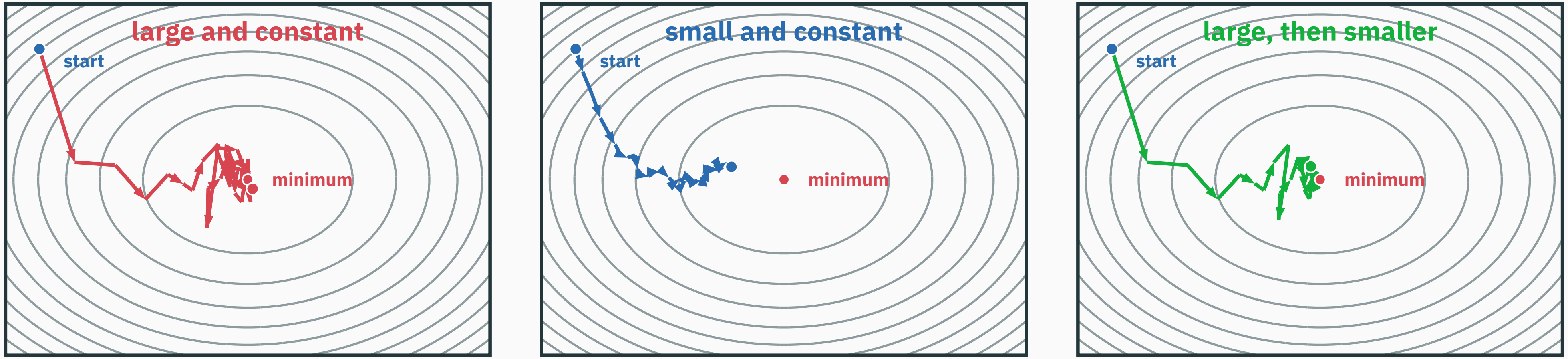
$$g_{B(\theta_t)} = \nabla L(\theta_t) + \varepsilon_B \qquad \theta_{t+1} = \theta_t - \eta_t g_{B(\theta_t)}$$

Far away, the gradient signal is large compared with the minibatch noise.

Near the minimum, the gradient signal can be smaller than the minibatch noise.

# The learning rate sets the size of the bounce

Q QUESTION




$\eta = .20$: arrives quickly, then jitters

$\eta = .04$: quieter, but slower

$.20 \to .04$: fast start, quiet finish

For the same noise term $|\varepsilon_B| = .5$:

$\boxed{\eta = .20 \to \eta|\varepsilon_B| = .10}$ $\boxed{\eta = .04 \to \eta|\varepsilon_B| = .02}$

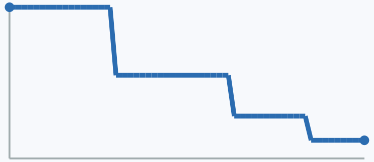
**A schedule keeps the useful large steps and reduces the late noise-driven moves.**

# Four useful decay rules



## Step decay

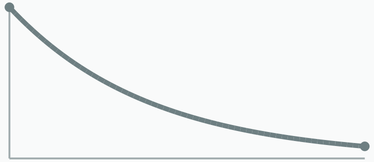
$$\eta_t = \eta_0 \gamma^{\lfloor \frac{t}{K} \rfloor}$$



**Use it when:** you know the milestones and want drops that are easy to see.

## Exponential decay

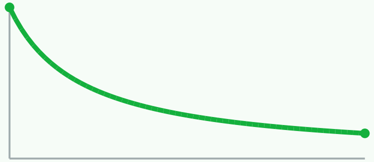
$$\eta_t = \eta_0 \exp(-kt)$$



**Use it when:** you want a steady percentage decrease with no abrupt jumps.

## Inverse-time decay

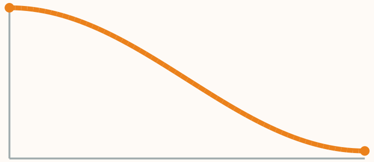
$$\eta_t = \frac{\eta_0}{1+kt}$$



**Use it when:** you want an early reduction followed by a long, gentle tail.

## Cosine decay $(0 \leq t \leq T)$

$$\eta_t = \eta_{\min} + \frac{\eta_0 - \eta_{\min}}{2}\left(1 + \cos\left(\pi \frac{t}{T}\right)\right)$$



**Use it when:** training ends at $T$.

**Feedback rule: reduce on plateau.** Use it when validation progress, not the clock, should trigger a drop. Patience prevents one noisy check from changing $\eta$.

# PyTorch: two scheduler patterns

I INTERACTIVE

```python
from torch.optim import lr_scheduler as lrs
optimizer = torch.optim.SGD(model.parameters(), lr=0.1)
```

## FIXED CLOCK: COSINE

```python
cosine = lrs.CosineAnnealingLR(
    optimizer,
    T_max=num_epochs,
    eta_min=1e-4,
)

for epoch in range(num_epochs):
    train_one_epoch(...)
    cosine.step()  # after optimizer updates
```

**Clock:** `T_max` counts calls to `cosine.step()`. Here, one call means one epoch.

## VALIDATION FEEDBACK: PLATEAU

```python
plateau = lrs.ReduceLROnPlateau(
    optimizer, mode="min",
    factor=0.1, patience=3,
)

for epoch in range(num_epochs):
    train_one_epoch(...)
    val_loss = validate(...)
    plateau.step(val_loss)
```

**Metric:** `factor=.1` makes each drop 10x smaller. `patience=3` waits before reacting to a short stall.

**The call tells you what drives the schedule: `step()` uses time; `step(val_loss)` uses validation.**

# PyTorch: one epoch, step by step

I INTERACTIVE

```
from torch.optim import lr_scheduler as lrs

optimizer = torch.optim.SGD(model.parameters(), lr=0.1)
scheduler = lrs.CosineAnnealingLR(
    optimizer,
    T_max=num_epochs,
    eta_min=1e-4,
)

for epoch in range(num_epochs):
    model.train()
    for x, y in train_loader:
        optimizer.zero_grad()          # 1
        y_hat = model(x)               # 2
        loss = loss_fn(y_hat, y)       # 3
        loss.backward()                # 4
        optimizer.step()               # 5

    scheduler.step()                   # 6
```

## WHAT EACH NUMBER DOES

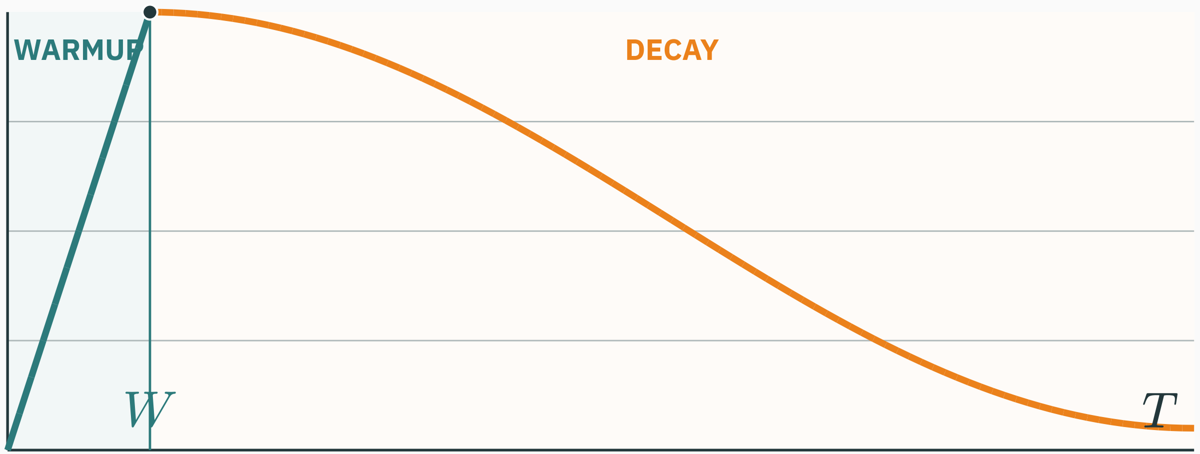
1. **Clear old gradients.** PyTorch adds gradients unless we reset them.
2. **Predict.** Run the minibatch through the model.
3. **Measure the error.** Compare the prediction with the target.
4. **Backpropagate.** Fill each parameter's `.grad`.
5. **Move the weights.** Use the current learning rate.
6. **Advance the schedule.** Set the rate for the next epoch.

One call per epoch means `T_max=num_epochs`. If you call the scheduler every batch, its clock must count batches instead.
**Timing:** `optimizer.step()` moves the model; `scheduler.step()` sets the rate for future moves.

**If validation should decide:** keep steps 1-5. After validation, use `val_loss = evaluate(model, val_loader)` and then `plateau.step(val_loss)`.

# Warm up, then decay

V VISUAL




Example: warm up for the first 12% of updates, then use cosine decay to the finish.

**Why it helps:** Early gradients can be unusually large. Since $|\Delta\theta_t| = \eta_t |g_t|$, a small $\eta_t$ keeps the first parameter changes small and reduces the risk of a loss spike or divergence.

### 1. Start small

$$\eta_t = \eta_{\text{peak}} \frac{t}{W}$$

Raise the rate gradually over the first $W$ updates.

### 2. Then decay

$$\eta_t \to \eta_{\min}$$

Once the early updates are stable, shrink the rate using the chosen rule.

**Look for this:** the loss spikes or becomes NaN when training starts at the full learning rate.

# What should you remember?

Q QUESTION

### Far from the minimum
The gradient signal is often strong. A large learning rate can make fast progress.

### Near the minimum
Minibatch noise can dominate. A smaller learning rate reduces the resulting motion.

### Fixed training budget
Use a time-based schedule such as cosine decay.

### Validation should decide
Use reduce on plateau after improvement stalls.

$$g_{B(\theta_t)} = \nabla L(\theta_t) + \varepsilon_B \qquad \Delta\theta_t = -\eta_t g_{B(\theta_t)}$$

**Large steps help early. Smaller steps give SGD more control near the minimum.**

Choose the simplest rule that matches the run. Add warmup only when startup is fragile.

# Sources and evidence

- MIT OCW · Lecture 25: Stochastic Gradient Descent: fast early progress, followed by bouncing near the solution.
- Fabian Pedregosa · The Stochastic Gradient Method: interactive 2D paths for constant and decreasing step sizes.
- Fabian Pedregosa · Gradient Descent: interactive contour plots showing sensitivity to step size.
- Andrew Ng · Learning Rate Decay: one contour picture and one numerical decay example.
- PyTorch · Learning-rate schedulers: scheduler call order and metric-based decay.